

THAARU Sepsis AI — Clinical Intelligence

Exploratory Data Analysis (EDA) Research Report • Generated: July 19, 2026

1. Executive Summary & Dataset Overview

This report outlines the clinical characteristics and temporal trajectories of the PhysioNet Sepsis Challenge 2019 dataset. The consolidated cohort includes files from both Training Set A and Training Set B, merging individual patient records into a standardized chronological database. This structured data foundation serves as the core training artifact for our Sepsis predictive models.

Metadata Parameter	Value
Dataset Source	PhysioNet Challenge 2019
Unique Patients Cohort	40,336
Total Hourly Records	1,552,210
Features (Vitals & Labs)	40
Stay Length (Mean / Median)	39.01h / 39.0h
Stay Length (Min / Max Range)	8h - 336h
Consolidated CSV Size	141.5 MB
Consolidated Parquet Size	15.7 MB (9x compressed)

2. Cohort Sepsis & Class Distribution

A critical challenge in Sepsis detection is class imbalance. Sepsis is a relatively rare clinical event at an hourly granularity. However, at the patient cohort level, a substantial subset of patients eventually develops sepsis, offering rich temporal sequences for ML training.

Category	Class Label	Count	Ratio (%)
Hourly Records	Non-Sepsis (0)	1,524,317	98.20%
Hourly Records	Sepsis (1)	27,893	1.80%
Patient Level	Never Septic	37,404.0	92.73%
Patient Level	Sepsis (Ever Septic)	2,932.0	7.27%

3. Clinical Distribution Figures

Below are the primary cohort distributions including patient ICU lengths of stay and sepsis class rates. These distributions illustrate the extreme skew at the hourly level (1.8% positive labels) contrasted with the patient cohort level (which contains approximately 7.2% septic patients).

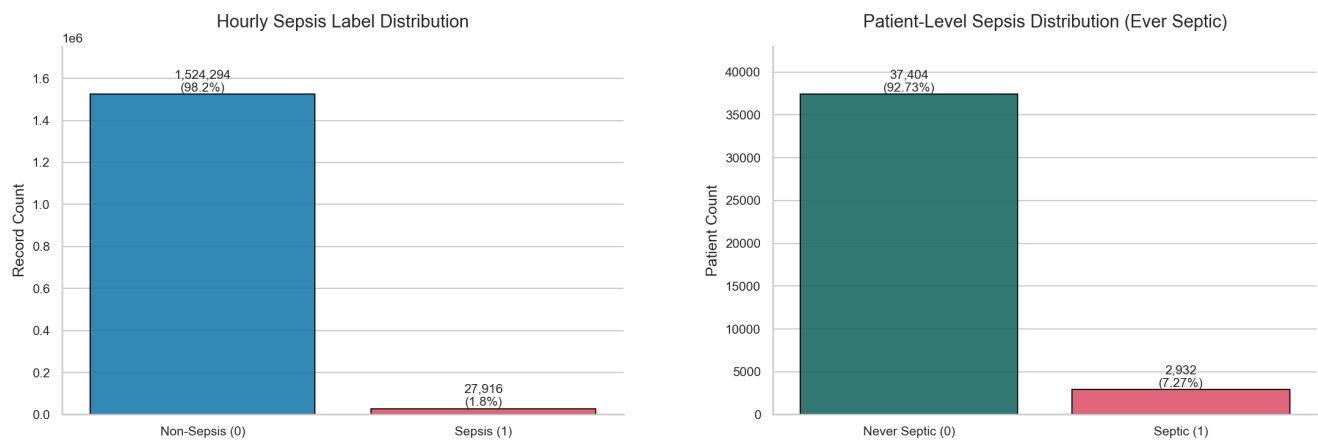
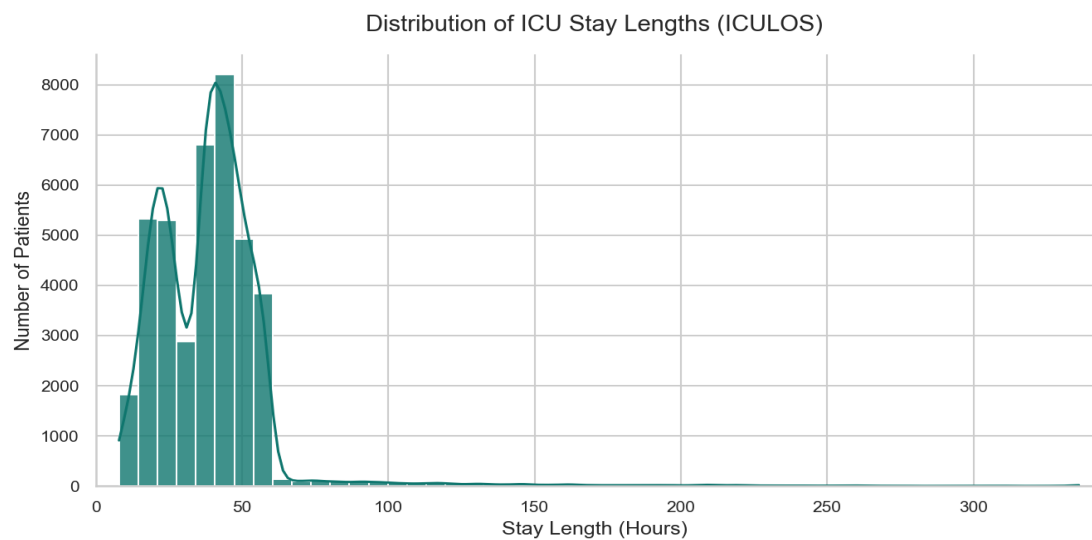
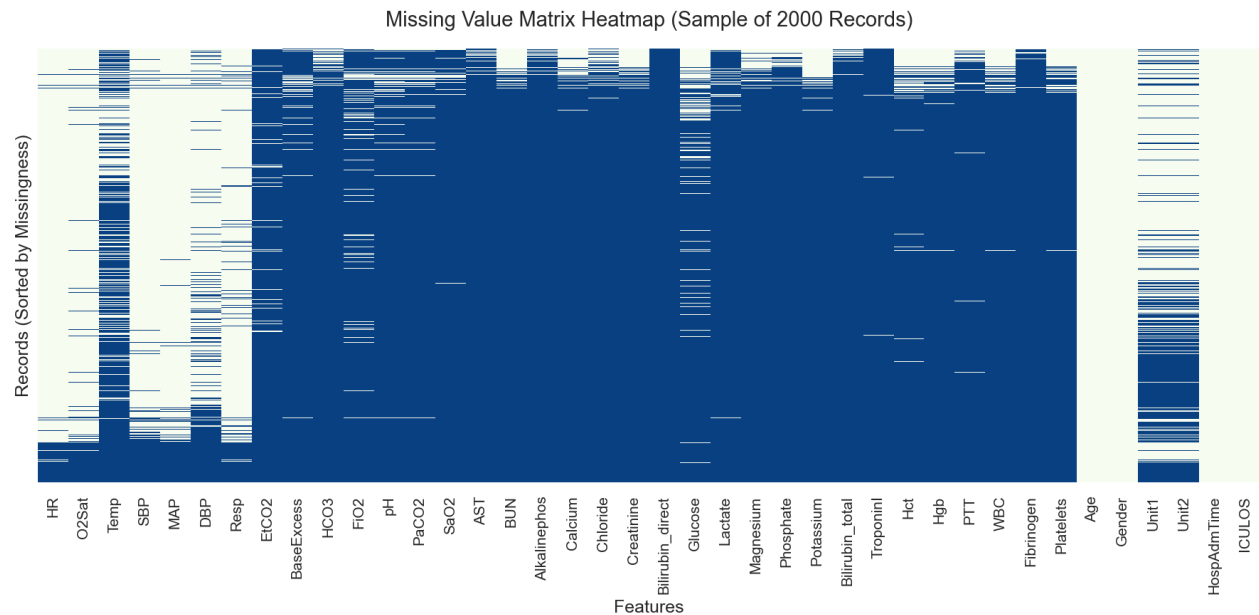


Figure 1: Distribution of ICU stay lengths (ICULOS)



4. Missing Data & Missingness Heatmap

Clinical datasets from real ICU environments contain extensive missingness because vital signs and labs are measured at discrete, non-uniform intervals. The heatmap below displays missingness patterns across features for a cohort sample. Vital signs (HR, O2Sat, Resp) are recorded frequently, whereas laboratory metrics (Lactate, Bilirubin, Creatinine) are highly sparse (often missing >90% of the time).



Feature Name	Null Record Count	Missingness Ratio (%)
Bilirubin_direct	1,549,220	99.81%
Fibrinogen	1,541,968	99.34%
TroponinI	1,537,429	99.05%
Bilirubin_total	1,529,069	98.51%
Alkalinephos	1,527,269	98.39%
AST	1,527,027	98.38%

5. Feature Correlations & Outliers

Understanding collinearity and impossible outliers guides model regularization and robust normalization. Highly correlated clusters exist among blood pressure readings (SBP, MAP, DBP) and hematology indicators (Hgb, Hct). Additionally, boxplots below isolate outlier ranges for physiological parameters.

Figure 2: Correlation heatmap of key clinical vitals & labs

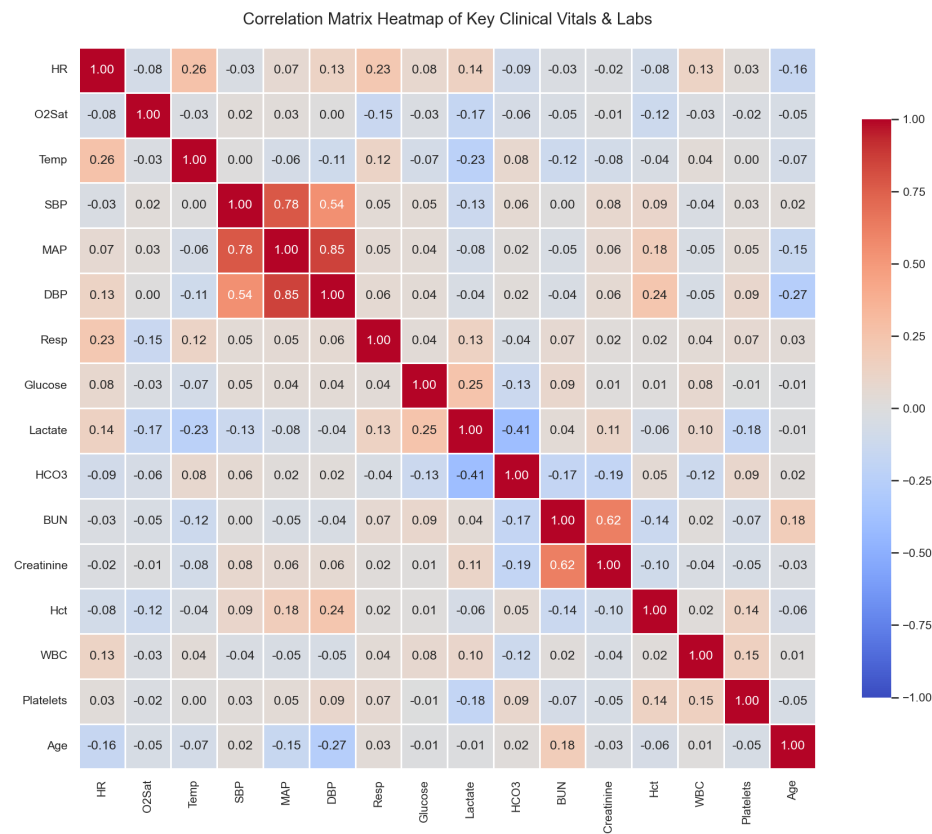
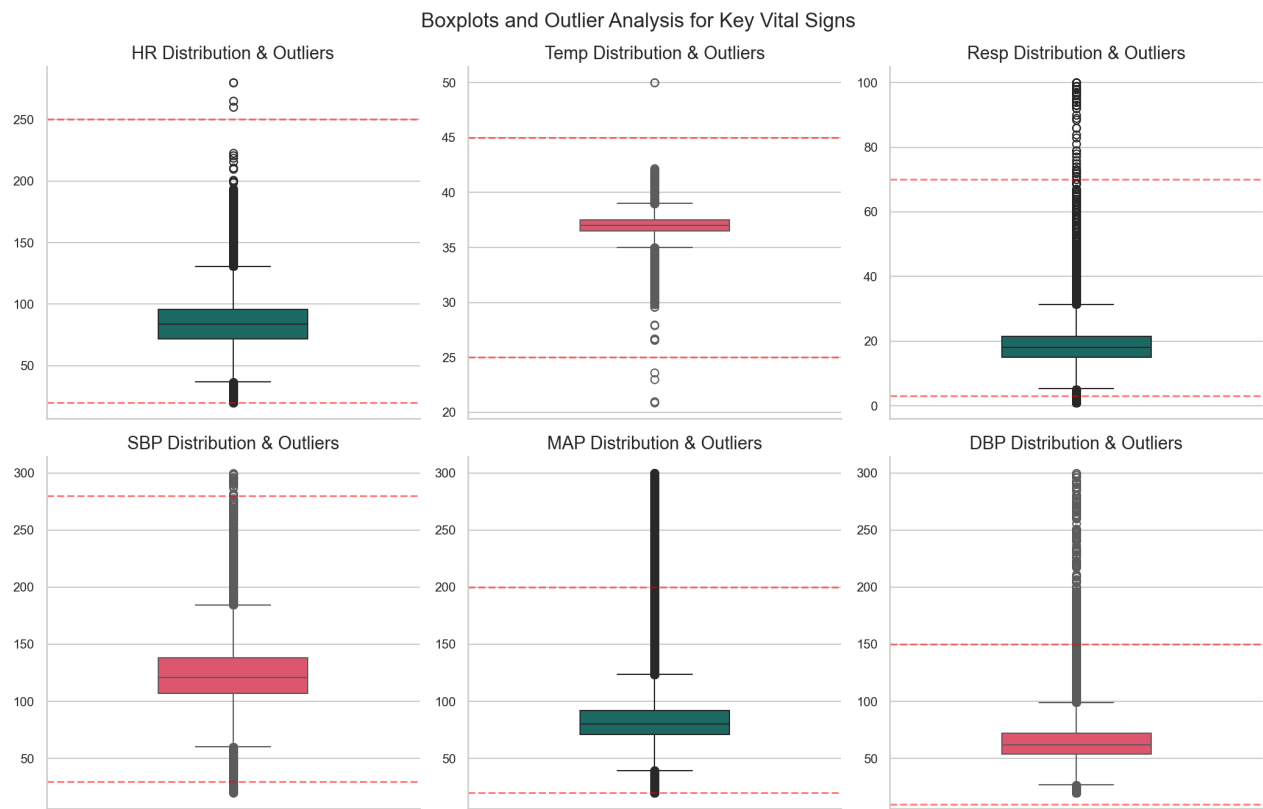
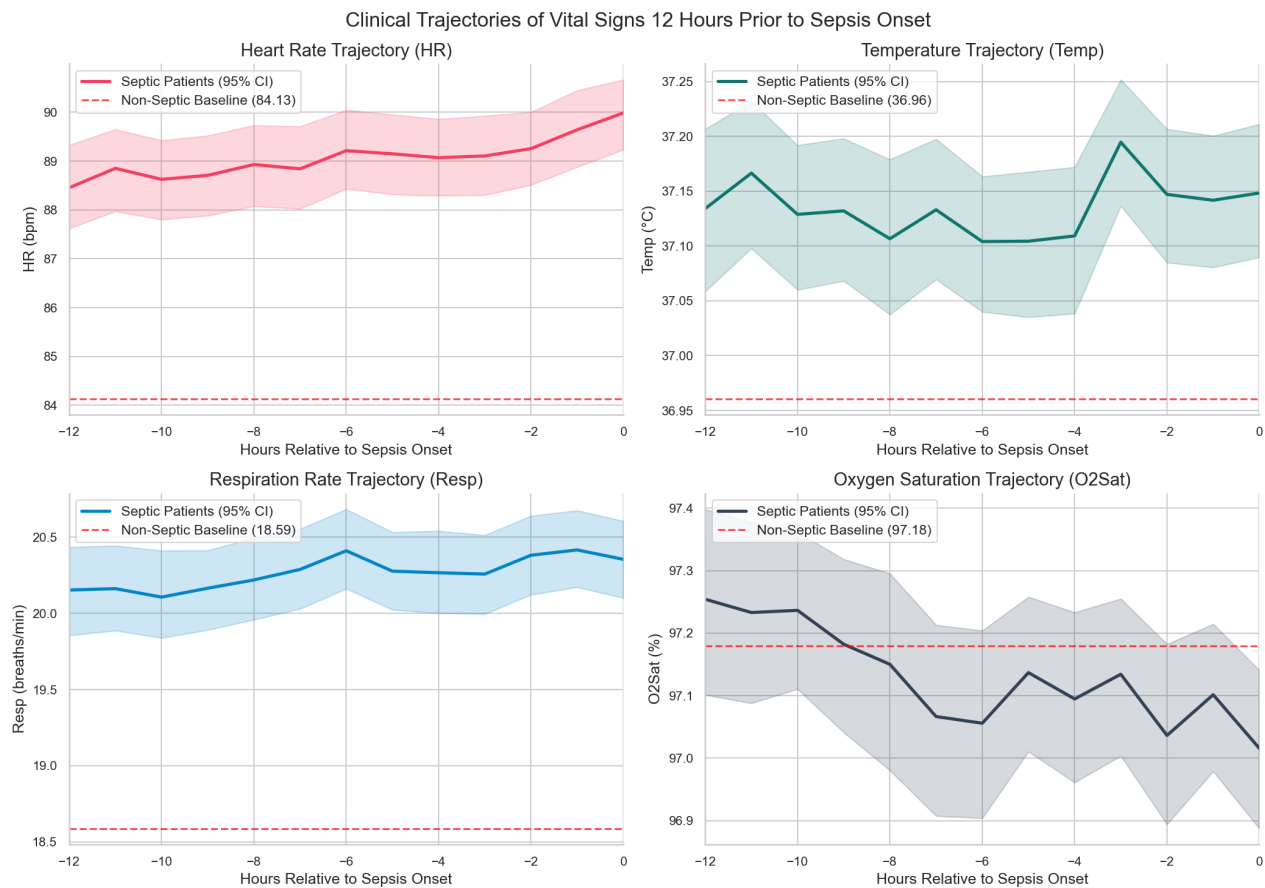


Figure 3: Outlier distributions of primary vitals



6. Physiological Trajectories Prior to Sepsis

By aligning septic patient timelines around their first hour of positive sepsis diagnosis, we extract the physiological trend trajectories. As shown below, during the 12 hours preceding diagnosis, there is a clear deterioration in vital signs: an upward drift in Heart Rate (HR) and Respiration Rate (Resp), accompanied by a decrease in Oxygen Saturation (O2Sat) and temperature instability. This trend strongly supports our proposed use of time-series models like LSTM to identify early sepsis indicators.



7. Preprocessing & Modelling Guidelines

Based on these empirical findings, we establish the following requirements for the ML pipeline:

- 1. Imputation:** Since lab values are sparse (>90% missing), we must employ forward-fill with patient-specific baselines rather than global means to avoid introducing bias.
- 2. Normalization:** Robust scaling is recommended for features like HR, SBP, and MAP because of statistical outliers.
- 3. Imbalance Mitigation:** Sepsis is highly imbalanced at the hourly level (1.8%). Loss functions must use class-weighted focal loss or downsampling techniques.